Lab Sheet 09

Title: Work with Query and Projection Operators.

Objective:

- Specify Conditions Using Query Operators.
- Specify AND, OR, NOT Conditions.

<u>Comparison</u>

Equal	{<key>: {$eg :<value>}}
Greater than Greater than equal	{<key>: {$gt:<value>}} {<key>: {$gte:<value>}}
Values in an array	{<key>: {$in:[<value1>, <value2>,……<valueN>]}}
Less than	{<key>: {$lt:<value>}}
Less than equal	{<key>: {$lte:<value>}}
Not equal	{<key>: {$ne:<value>}}

<u>Logical</u>

$and db.collectionName.find({ $and: [{key1:value1}, { key2:value2}] })

$or db.collectionName.find({ $or: [{key1:value1}, { key2:value2}] })

$not db.collectionName.find({ $not: [{key1:value1}, { key2:value2}] })

Database: CompanyData

Collection: inventory

Item	Quantity	Status	Category	Supplier	Price	Location
Note Book	25	A	Stationery	ABC Traders	45.00	Shelf-A
Journal	50	B	Stationery	XYZ Ltd	85.00	Shelf-B
Paper	100	A	Office Use	ABC Traders	15.00	Shelf-C
Article	36	C	Educational	LearnFast	120.0	Shelf-A
Post Card	81	D	Gifting	PrintMate	25.00	Shelf-B
Pen	60	A	Stationery	ABC Traders	10.00	Shelf-C
Eraser	20	A	Stationery	XYZ Ltd	5.00	Shelf-A
Marker	75	B	Office Use	WriteIt	30.00	Shelf-D
File	90	B	Office Use	ABC Traders	55.00	Shelf-E
Ruler	10	A	Stationery	Tools & Co	8.00	Shelf-A
Envelope	45	C	Mailing	XYZ Ltd	20.00	Shelf-B
Highlighter	55	A	Stationery	LearnFast	28.00	Shelf-C

1. **Select all the documents in the inventory collection where the status equals “A”.**

    db.inventory.find({ status: "A" })

2. **Return all the documents where the quantity is less than 30.**

    db.inventory.find({ quantity: { $lt: 30 } })

3. **Return all the documents where the status equals “A” and quantity is less than 30.**

```
db.inventory.find({
$and: [
   { status: "A" },
   { quantity: { $lt: 30 } }
 ]
})
```

4. **Select all the documents where the quantity is greater than 50 and the item is Post Card.**

```
db.inventory.find({
 $and: [
   { quantity: { $gt: 50 } },
   { item: "Post Card" }
 ]
})
```

5. **Select the documents where the quantity is not equal to 100.**

```
db.inventory.find({ quantity: { $ne: 100 } })
```

6. **Select the documents where either status is "A" or the quantity is not greater than 70.**

```
db.inventory.find({
    $or: [
      { status: "A" },
      { quantity: { $lte: 70 } }
    ]
   })
```

<u>Exercise</u>

Database: LibraryDB **Collection**: libraryBooks

BookID	Title	Author	Category	Copies	PublishedYear	Status
B001	Introduction to Databases	C.J. Date	Database	5	2018	Available
B002	Learning Python	Mark Lutz	Programming	2	2019	Borrowed
B003	MongoDB Basics	Kristina Chodorow	Database	8	2020	Available
B004	Clean Code	Robert C. Martin	Programming	4	2008	Available
B005	Web Development Handbook	Jon Duckett	Web	3	2017	Borrowed
B006	JavaScript: The Good Parts	Douglas Crockford	Web	7	2009	Available
B007	React Up & Running	Stoyan Stefanov	Web	1	2021	Borrowed
B008	Advanced Java	Herbert Schildt	Programming	6	2016	Available
B009	Cloud Computing Concepts	Rajkumar Buyya	Cloud	4	2022	Borrowed
B010	Node.js in Action	Mike Cantelon	Programming	3	2015	Available
B011	Fundamentals of AI	Stuart Russell	AI	2	2020	Available
B012	Machine Learning Basics	Tom Mitchell	AI	5	2021	Borrowed

```
// Use Database
use LibraryDB

// Insert Data into collection "libraryBooks"
db.libraryBooks.insertMany([
```

```
{
  BookID: "B001",
  Title: "Introduction to Databases",
  Author: "C.J. Date",
  Category: "Database",
  Copies: 5,
  PublishedYear: 2018,
  Status: "Available"
},
{
  BookID: "B002",
  Title: "Learning Python",
  Author: "Mark Lutz",
  Category: "Programming",
  Copies: 2,
  PublishedYear: 2019,
  Status: "Borrowed"
},
{
  BookID: "B003",
  Title: "MongoDB Basics",
  Author: "Kristina Chodorow",
  Category: "Database",
  Copies: 8,
  PublishedYear: 2020,
  Status: "Available"
},
{
  BookID: "B004",
  Title: "Clean Code",
  Author: "Robert C. Martin",
  Category: "Programming",
  Copies: 4,
  PublishedYear: 2008,
  Status: "Available"
},
{
  BookID: "B005",
  Title: "Web Development Handbook",
  Author: "Jon Duckett",
  Category: "Web",
  Copies: 3,
  PublishedYear: 2017,
  Status: "Borrowed"
},
```

```
{
  BookID: "B006",
  Title: "JavaScript: The Good Parts",
  Author: "Douglas Crockford",
  Category: "Web",
  Copies: 7,
  PublishedYear: 2009,
  Status: "Available"
},
{
  BookID: "B007",
  Title: "React Up & Running",
  Author: "Stoyan Stefanov",
  Category: "Web",
  Copies: 1,
  PublishedYear: 2021,
  Status: "Borrowed"
},
{
  BookID: "B008",
  Title: "Advanced Java",
  Author: "Herbert Schildt",
  Category: "Programming",
  Copies: 6,
  PublishedYear: 2016,
  Status: "Available"
},
{
  BookID: "B009",
  Title: "Cloud Computing Concepts",
  Author: "Rajkumar Buyya",
  Category: "Cloud",
  Copies: 4,
  PublishedYear: 2022,
  Status: "Borrowed"
},
{
  BookID: "B010",
  Title: "Node.js in Action",
  Author: "Mike Cantelon",
  Category: "Programming",
  Copies: 3,
  PublishedYear: 2015,
  Status: "Available"
},
```

```
{
  BookID: "B011",
  Title: "Fundamentals of AI",
  Author: "Stuart Russell",
  Category: "AI",
  Copies: 2,
  PublishedYear: 2020,
  Status: "Available"
},
{
  BookID: "B012",
  Title: "Machine Learning Basics",
  Author: "Tom Mitchell",
  Category: "AI",
  Copies: 5,
  PublishedYear: 2021,
  Status: "Borrowed"
}
])
```

1. **Select all books in the "Programming" category.**

```
db.LibraryBooks.find({Category: "Programming"})
```

2. **Return all books that have fewer than 4 copies.**

```
db.LibraryBooks.find({Copies: { $lt: 4}})
```

3. **Select all books that are "Available" and published after 2015.**

```
db.LibraryBooks.find({
  $and: [{ Status: "Available"},{ PublishedYear: { $gt: 2015}}]
})
```

4. **Find all books with the word "Basics" in the title.**

```
db.LibraryBooks.find({Title: { $regex: "Basics", $options: "i"}})
```

5. **Return books that were published before 2010 or have more than 5 copies.**

```
db.LibraryBooks.find({
  $or: [{ PublishedYear: { $lt: 2010}},{ Copies: { $gt: 5}}]
})
```

6. **Select books where the category is not "Web".**

```
db.LibraryBooks.find({Category: { $ne: "Web"}})
```

7. **Return books where the author is either "Mark Lutz" or "Robert C. Martin".**

```
db.LibraryBooks.find({Author: {$in: ["Mark Lutz","Robert C. Martin"]}})
```

8. **Find all books with "Borrowed" status and fewer than 3 copies.**

```
db.libraryBooks.find({
  $and: [{ Status: "Borrowed"},{ Copies: {$lt: 3}}]
})
```

9. **Return all books with PublishedYear between 2016 and 2020 (inclusive).**

```
db.libraryBooks.find({
  PublishedYear: {$gte: 2016,$lte: 2020}
})
```